



SIMATS ENGINEERING
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CSA0927 - PROGRAMMING IN JAVA

**A Smart Event Management Framework for Real-Time Crowd
Monitoring, Secure Entry, and Predictive Analytics**

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INTRODUCTION

- Large-scale events often face challenges such as overcrowding, long entry queues, security risks, and inefficient event planning.
- Traditional event management systems rely on manual processes, making them slow and error-prone.
- Smart technologies like QR-based ticketing, AI-driven crowd monitoring, and predictive analytics improve event operations.
- This framework integrates automation, real-time monitoring, and intelligent decision-making.
- It enhances visitor experience, organizer efficiency, and overall event safety.



Problem Statement



- Manual ticket verification causes long waiting times.
- Lack of real-time crowd monitoring increases safety risks.
- Difficult to predict attendance and resource requirements.
- Inefficient communication between organizers and security teams.
- Limited insights into event performance after completion.
- Existing systems do not provide integrated planning, monitoring, and analytics.



Objectives



- Develop a centralized smart event management platform.
- Enable secure QR-code based ticket verification.
- Monitor crowd density using AI and real-time analytics.
- Predict attendance and optimize resource allocation.
- Improve visitor safety through intelligent alerts.
- Provide interactive dashboards for event organizers.
- Enhance decision-making using predictive analytics.



Proposed System



- **Proposed System**

- The proposed system consists of four intelligent modules:
- Event Planning and Scheduling
- QR-Based Ticket Generation and Validation
- AI-Based Crowd Monitoring
- Predictive Analytics Dashboard

- **Features**

- Automated ticket management
- Secure QR authentication
- Real-time crowd monitoring
- Capacity alerts
- Attendance prediction
- Interactive dashboards

Literature Review

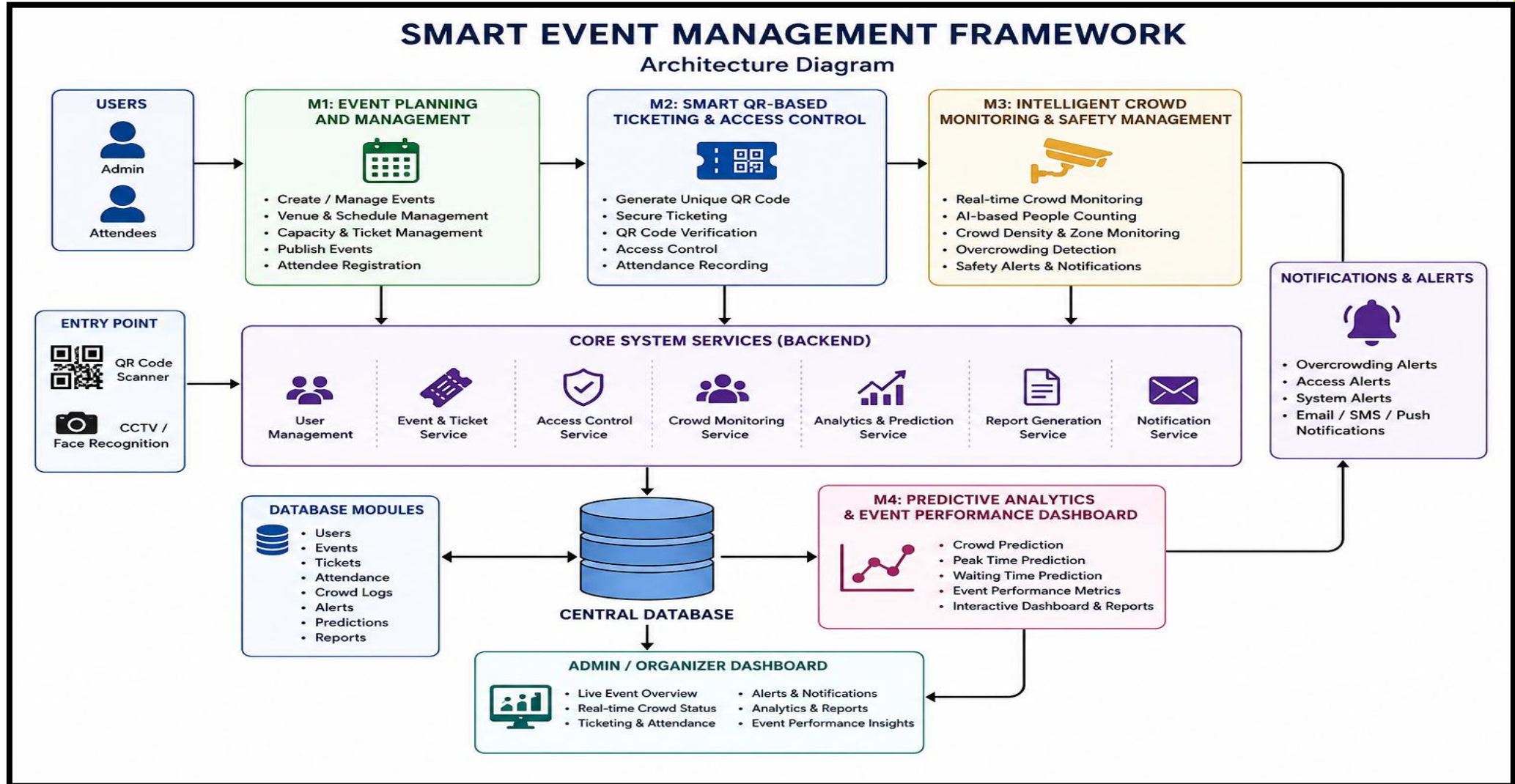
S. No	Year	Author	Title	Journal	Methodology	Drawbacks
1	2025	Darsena et al.	Sensing Technologies for Crowd Management	IEEE Review	IoT, Crowd Sensing	Limited to transportation systems
2	2024	Zhou et al.	Understanding Crowd Behaviors in a Social Event	arXiv	WiFi Sensing, Data Mining	No ticketing or access control
3	2024	Ameen et al.	Advancements in Crowd-Monitoring Systems	arXiv	AI, Computer Vision	High computational cost
4	2023	collins	Event Management System with QR Code-Based Check-in	SJAIBT	QR Code, Cloud Computing	Limited predictive analytics
5	2025	Roy Choudhury et al.	Drishti AI-Event Guardian	arXiv	YOLOv8, Deep Learning	Focuses mainly on crowd monitoring



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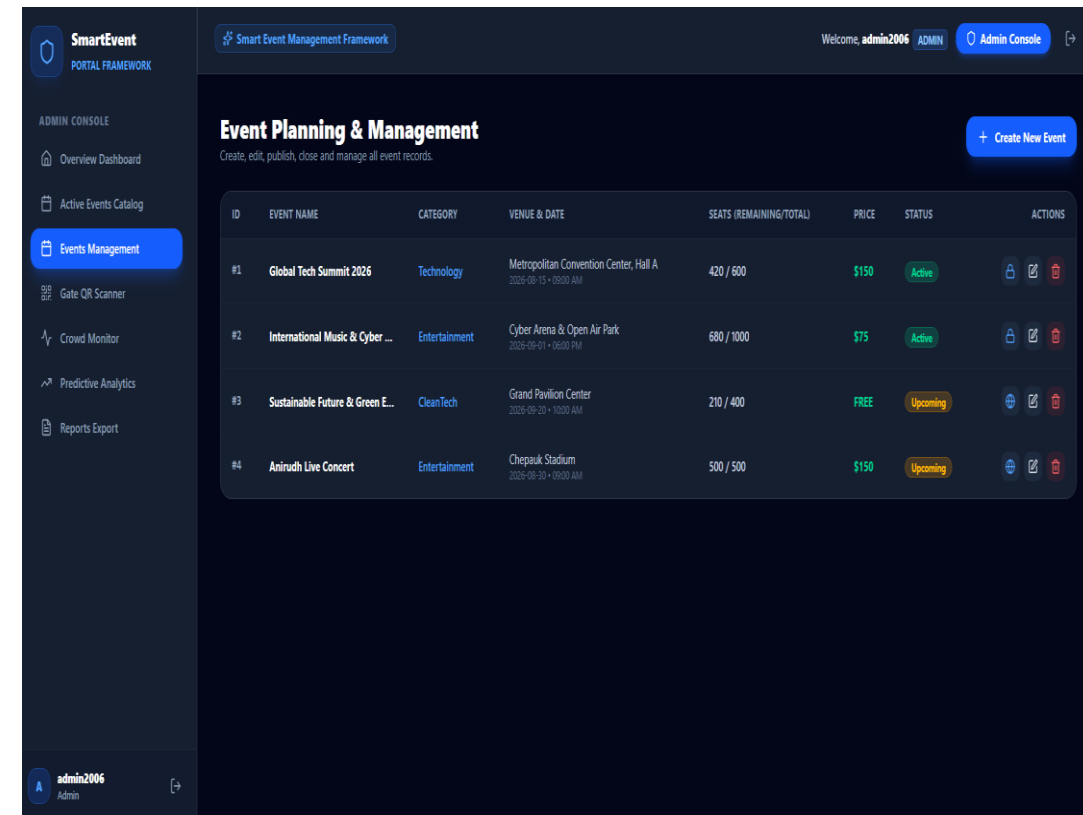


System Architecture



Module 1 – Event Planning and Management

- Creates and schedules events with venue, date, and time details.
- Manages event resources such as staff, equipment, and vendors.
- Organizes event timelines and activity schedules.
- Allocates budgets and tracks event-related expenses.
- Sends notifications and updates to organizers and participants.
- Maintains centralized event information for smooth coordination.



The screenshot displays the SmartEvent Portal Framework interface. The top navigation bar includes the SmartEvent logo, a 'Smart Event Management Framework' link, and user information: 'Welcome, admin2006 ADMIN' and an 'Admin Console' button. The left sidebar lists the 'ADMIN CONSOLE' with options: Overview Dashboard, Active Events Catalog, Events Management (highlighted), Gate QR Scanner, Crowd Monitor, Predictive Analytics, and Reports Export. The main content area is titled 'Event Planning & Management' with a subtitle 'Create, edit, publish, close and manage all event records.' and a '+ Create New Event' button. Below this is a table listing four events:

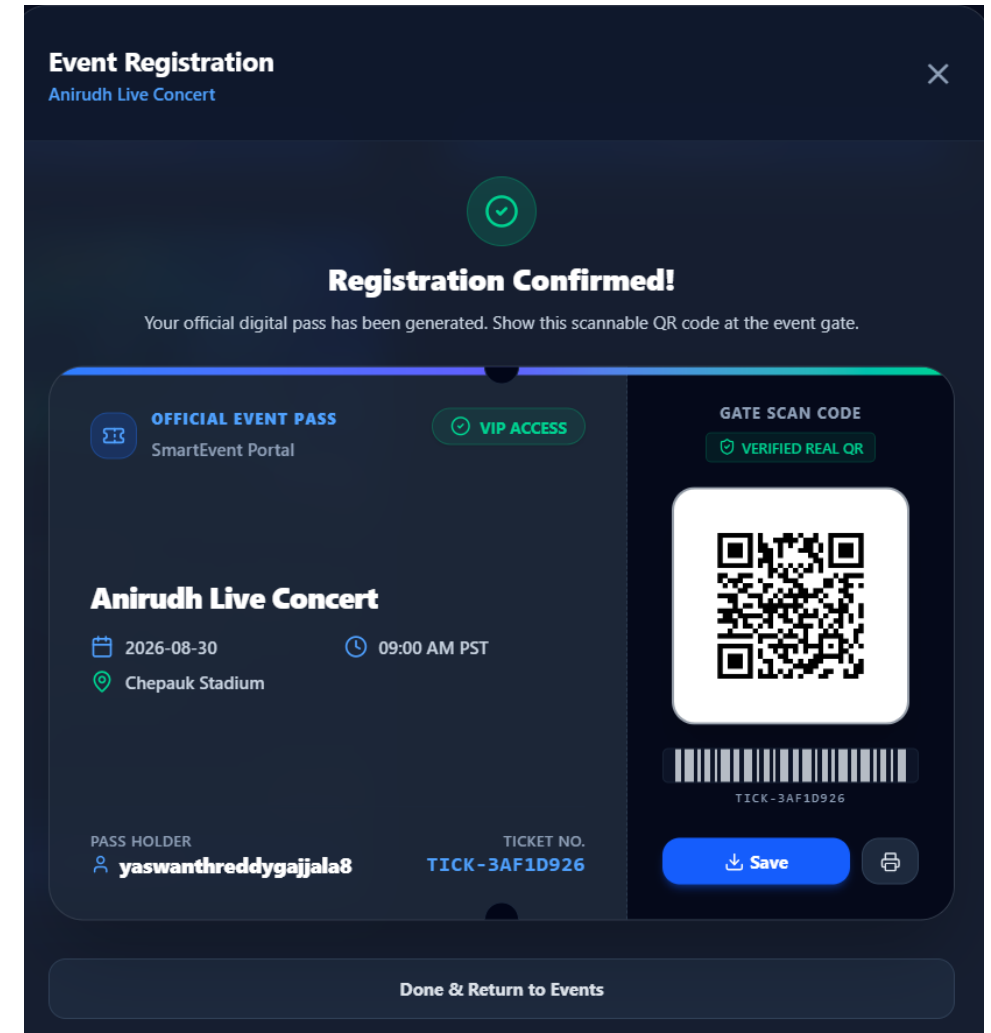
ID	EVENT NAME	CATEGORY	VENUE & DATE	SEATS (REMAINING/TOTAL)	PRICE	STATUS	ACTIONS
#1	Global Tech Summit 2026	Technology	Metropolitan Convention Center, Hall A 2026-09-15 • 09:00 AM	420 / 600	\$150	Active	[Lock] [Edit] [Delete]
#2	International Music & Cyber ...	Entertainment	Cyber Arena & Open Air Park 2026-09-01 • 06:00 PM	680 / 1000	\$75	Active	[Lock] [Edit] [Delete]
#3	Sustainable Future & Green E...	CleanTech	Grand Pavilion Center 2026-09-01 • 10:00 AM	210 / 400	FREE	Upcoming	[Lock] [Edit] [Delete]
#4	Anirudh Live Concert	Entertainment	Chepauk Stadium 2026-09-20 • 08:00 AM	500 / 500	\$150	Upcoming	[Lock] [Edit] [Delete]

The bottom left corner shows the user profile 'admin2006 Admin'.

Module 2: Smart QR-Based Ticketing and Access Control



- Generates unique and secure QR-code tickets for attendees.
- Validates QR codes at entry points for fast authentication.
- Prevents duplicate entries and detects fraudulent tickets.
- Tracks real-time attendance and entry/exit records.
- Reduces waiting time through automated access control.
- Enhances security with digital verification and monitoring.



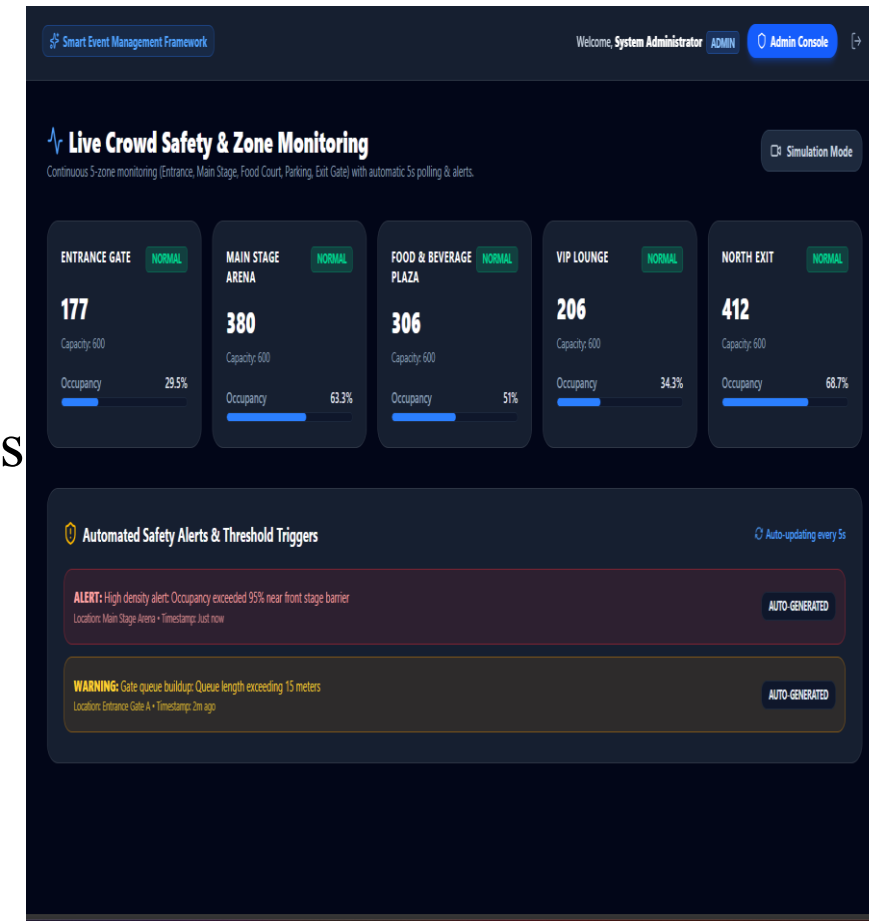


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Module 3: Intelligent Crowd Monitoring and Safety Management



- Monitors crowd density using AI-enabled cameras and sensors.
- Detects overcrowding and unusual crowd movements in real time.
- Generates instant safety alerts during emergency situations
- Assists security personnel with live crowd status updates.
- Visualizes crowd distribution through heatmaps and dashboards.
- Improves visitor safety by preventing congestion and accidents.





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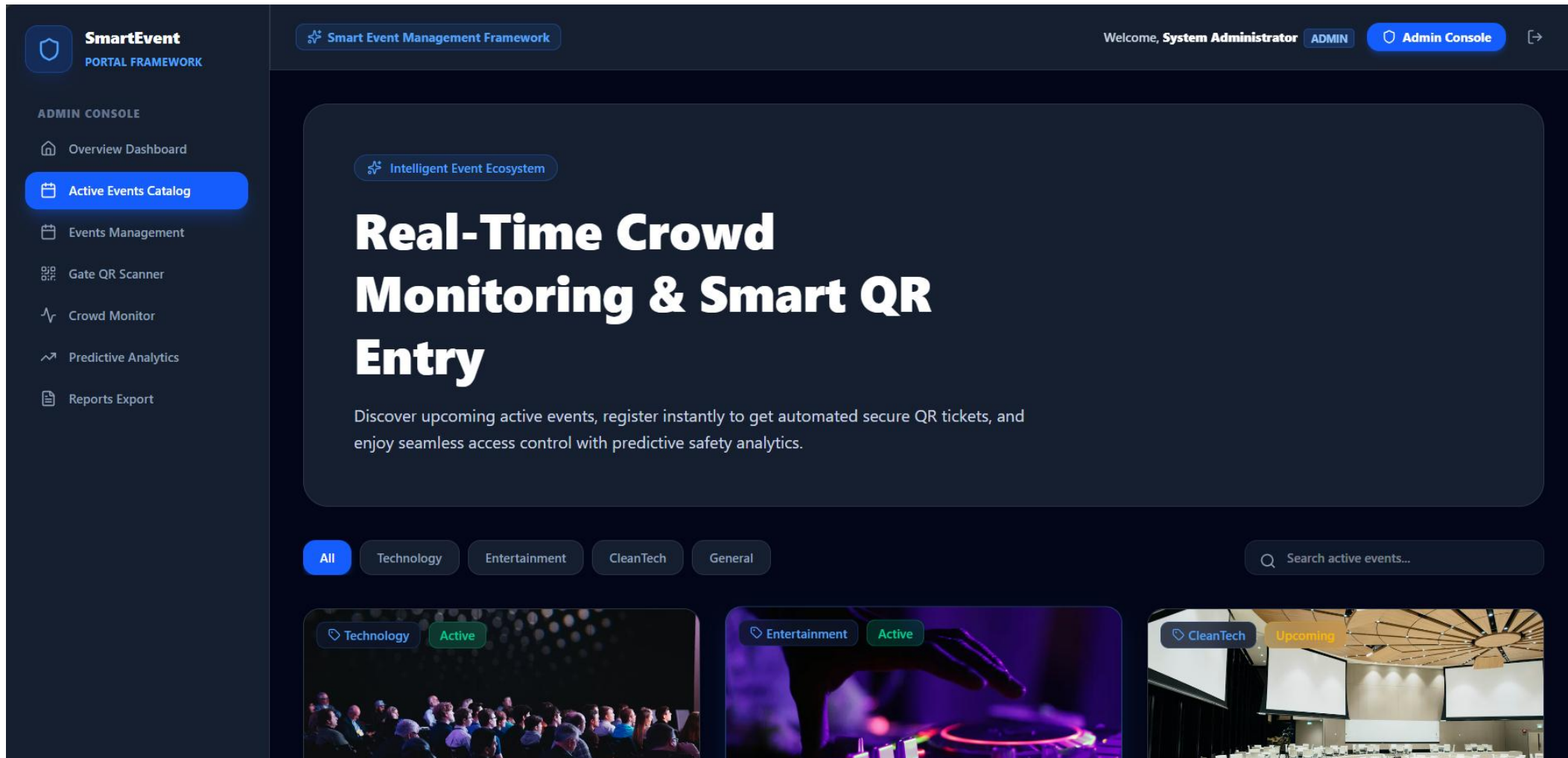
Module 4: Predictive Analytics and Event Performance Dashboard



- Analyzes historical and real-time event data.
- Predicts attendance, resource requirements, and revenue trends.
- Provides interactive dashboards with key performance indicators (KPIs).
- Evaluates visitor engagement and event success metrics.
- Supports data-driven decision-making for future events.
- Generates detailed reports to improve event planning and management.



Final output





Challenges & Limitations



- **Challenges**

- Large-scale real-time data processing
- Camera placement limitations
- Network connectivity issues
- Data synchronization
- Privacy concerns

- **Limitations**

- Requires internet connectivity
- Camera quality affects crowd monitoring accuracy
- AI prediction depends on historical data quality
- High initial deployment cost



Future Scope



- Facial recognition entry system
- IoT sensor integration
- Drone-based crowd surveillance
- Smart parking management
- AI chatbot support
- Multi-event management platform
- Digital payment integration
- Cloud-based scalable deployment



Conclusion



- The proposed framework integrates planning, ticketing, crowd monitoring, and analytics into a single smart platform.
- AI and QR technologies improve operational efficiency and visitor safety.
- Predictive analytics supports better planning and decision-making.
- The system reduces manual effort and enhances the overall event experience.
- It provides a scalable solution suitable for concerts, exhibitions, conferences, and public gatherings.



References



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THANK YOU